

Assignment 1

Deadline: 4 July 2026 11:30 PM

Submission link: <https://forms.gle/6NsNaBMpeYsDZqK38>

Category: A (Answer any three)

- 1) Suppose a line segment starts at (120, 23) and ends at (423, 428). If we draw this problem using the DDA algorithm, determine the number of times x will be increased. How many pixels will be needed to draw this line?
- 2) Which problems persist in the DDA algorithm and how can it be solved?
- 3) Why is the value of y incremented/decremented by 1 instead of x, when the gradient is < -1 or > 1 ?
- 4) Do you think the Mid Point Line Drawing Algorithm is better than DDA? Provide reasoning.
- 5) What is aliasing? Why don't we increase resolution to fight aliasing?
- 6) Explain unweighted Area sampling antialiasing technique.
- 7) Write some difference between vector and raster display. What is frame buffer? Which display uses frame buffer?

Category: B (Answer any two)

- 8) Using the Digital Differential Analyzer (DDA) algorithm, find out the pixels between the lines whose endpoints are given:
 - a. (5, 16) to (13, 10)
 - b. (0, 0) to (4, 6)
- 9) There is a line with the equation, $y = -5x - 20$. The line segment starts from where it cuts the y-axis and ends at where it cuts the x-axis. Find the first 8 pixels using the Midpoint Line Drawing Algorithm.
- 10) Suppose a line segment has the equation $\frac{x}{7} - \frac{y}{12} = 5$, and the starting point is at $y = 12$ and the ending point is at $y = 0$. Find the first 6 pixels using the Midpoint Line Algorithm.